QUANTUM ERROR CORRECTION (QEC) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 31, 2026

Diagram-based questions on Quantum Error Correction (QEC)

Question 1

10.0 points

Analyze the circuit shown below of a stabilizer code. Identify how the syndrome measurement helps in error detection and explain the role of stabilizers in quantum error correction.

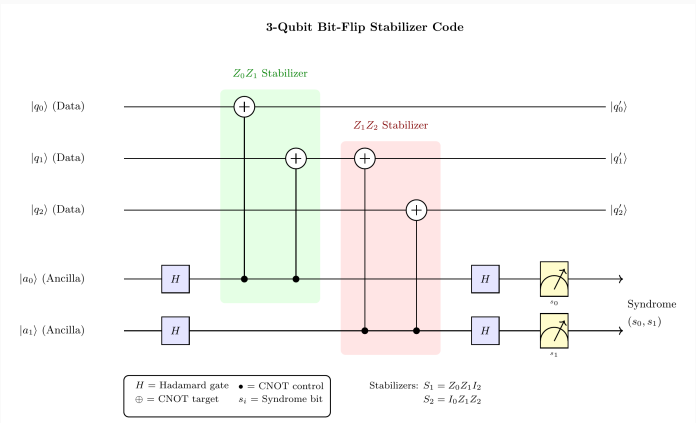

Fig. 1. A quantum circuit for a stabilizer code. Include qubits represented as horizontal lines, with quantum gates such as Hadamard (H), CNOT, and measurement gates. Show connections between qubits for CNOT gates, and indicate the measurement process for syndrome detection. Label the qubits and gates appropriately to represent a typical stabilizer code setup in quantum error correction.

Question 2

10.0 points

Examine the circuit shown below, which illustrates the 3-qubit bit-flip code along with its corresponding error detection circuit. Explain the error correction process and identify the role of each circuit component in detecting a single bit-flip error.

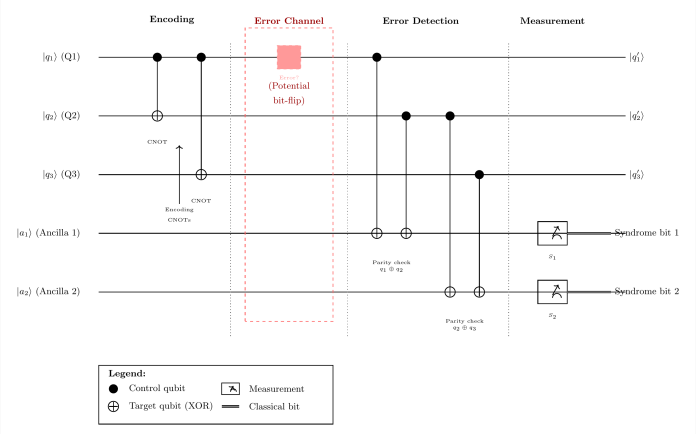

Fig. 2. A 3-qubit bit-flip code circuit with its corresponding error detection circuit. The diagram includes three qubits labeled Q1, Q2, and Q3. There are CNOT gates for encoding and error detection, with control and target qubits clearly marked. Measurement components are included to show how errors are detected. Each component is labeled to indicate its role in detecting a single bit-flip error. IMPORTANT correction needed: The diagram includes a "Syndrome Table". Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: The diagram is missing required labels. The diagram is missing labels for the ancilla qubits. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 3

10.0 points

Examine the surface code lattice in the diagram below. Describe how logical qubits are encoded on the lattice and discuss how the layout is used for error correction.

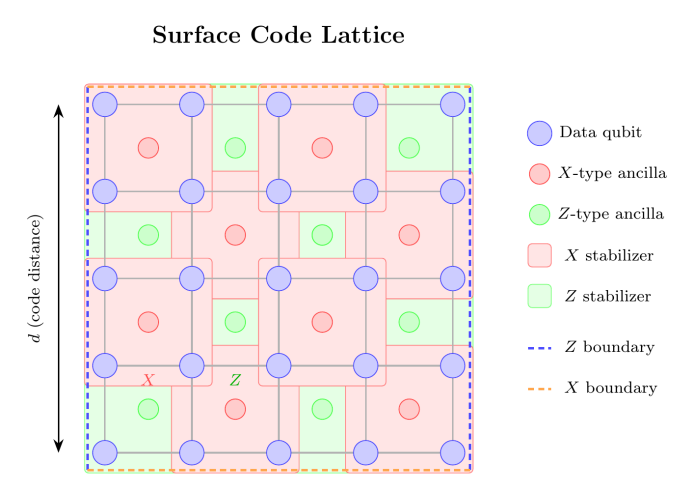

Fig. 3. A surface code lattice diagram showing a grid of qubits. Each qubit is represented as a circle. The lattice includes both data qubits and ancilla qubits. Data qubits are arranged in a grid pattern, while ancilla qubits are positioned between them to measure stabilizers. Highlight the stabilizer operators by drawing plaquettes around groups of qubits, indicating X and Z stabilizers. Label the logical qubits and show how they are encoded across the lattice. Include arrows or lines to represent error correction paths. IMPORTANT correction needed: The diagram explicitly draws and labels the logical. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: The diagram is missing required labels. The diagram is missing the required labels for the. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 4

10.0 points

Examine the diagram below illustrating fault-tolerant gate teleportation. Discuss how teleportation is utilized to achieve fault-tolerance in quantum gates and what protections it offers against errors.

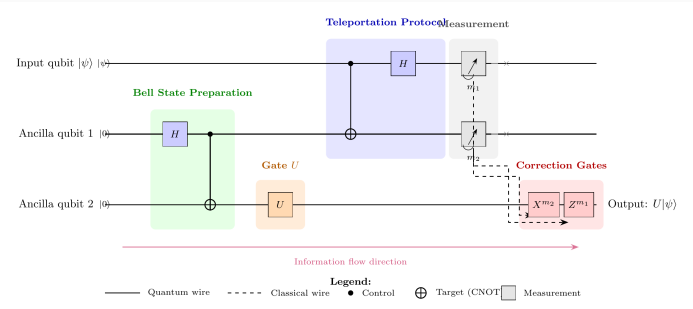

Fig. 4. A diagram illustrating fault-tolerant gate teleportation in quantum computing. Include qubits represented as horizontal lines, with quantum gates (such as CNOT, Hadamard) and measurement symbols along these lines. Show the initial state of the qubits, the teleportation process, and how errors are corrected. Label each component clearly, including qubits, gates, and measurement outcomes, to demonstrate how teleportation achieves fault-tolerance.

Question 5

10.0 points

Interpret the threshold theorem in the diagram below, which illustrates error rates in quantum error correction. Explain how the error rate threshold is defined and its significance in achieving fault-tolerant quantum computation.

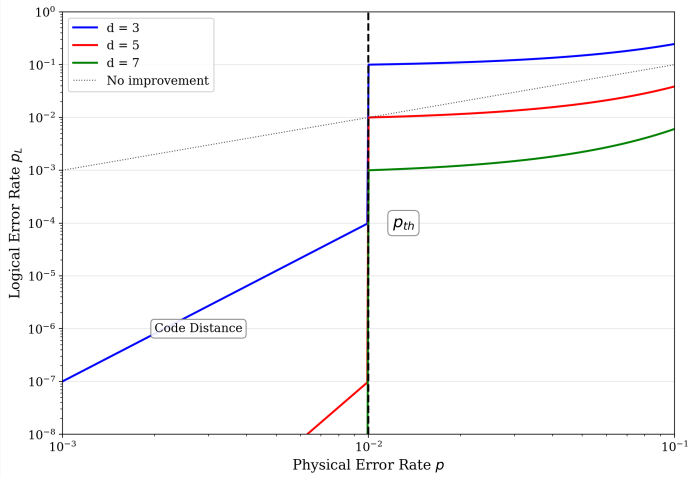

Fig. 5. CRITICAL: The previous diagram showed the wrong type. The diagram shows the logical. Regenerate from scratch showing the CORRECT diagram type. CRITICAL: The previous diagram showed the wrong type. The diagram shows a single curve where the logical error. Regenerate from scratch showing the CORRECT diagram type. A plot illustrating error rates in quantum error correction, showing the error rate threshold. The x-axis represents the error rate, and the y-axis represents the logical error rate. The plot includes a curve that shows how the logical error rate decreases as the physical error rate decreases, with a marked threshold line indicating the error rate threshold for fault-tolerant quantum computation.